Modeling with partial differential equations

Exercises – Session 2: Laplace and Poisson equations

Exercise 1: Another proof of the maximum principle in dimension 2

Let $f \in \mathcal{C}^2(\mathbb{R}^2)$ be harmonic on $\mathbb{R}^2$, i.e., $\Delta f = 0$ on $\mathbb{R}^2$. Let $\mathcal{D} = \{(x,y) \in \mathbb{R}^2 \,|\, x^2 + y^2 < r^2\}$ be the open disk of radius $r > 0$ and $\mathcal{C} = \partial\mathcal{D}$. Let $p \in \mathbb{N} \setminus \{0\}$. Let us define the function f_p on $\mathbb{R}^2$ by

$$f_p(x,y) = f(x,y) + \frac{x^2+y^2}{p}. \tag{1}$$

Question 1 Show that there exists $(a_p, b_p) \in \overline{\mathcal{D}}$ such that

$$f_p(a_p, b_p) = \max_{\overline{\mathcal{D}}} f_p(x,y). \tag{2}$$

Solution : *f_p is continuous on the compact set $\overline{\mathcal{D}}$.*

Question 2 Show that $(a_p, b_p) \in \mathcal{C}$.

Solution : *Assume that $(a_p, b_p) \in \mathcal{D}$. Then, $v : x \mapsto f_p(x, b_p)$ has a maximum in a segment $[a_p - \epsilon, a_p + \epsilon]$ for some $\epsilon > 0$ such that $[a_p - \epsilon, a_p + \epsilon] \subset \mathcal{D}$. Therefore, we must have*

$$v''(a_p) = \frac{\partial^2 f_p}{\partial x^2}(a_p, b_p) \leq 0.$$

Similarly, we have

$$\frac{\partial^2 f_p}{\partial y^2}(a_p, b_p) \leq 0. \tag{3}$$

After summing the two previous inequalities,

$$\frac{4}{p} \leq 0,$$

which is absurd. Notice that this argument cannot be used on f directly: $\Delta f \leq 0$ is not in contradiction with the assumption on f ($\Delta f = 0$).

Second method (second-order Taylor expansion and Hessian matrix). *Let $H_p(a_p, b_p)$ denote the Hessian matrix of f_p at (a_p, b_p):*

$$H_p(a_p, b_p) = \begin{pmatrix} \dfrac{\partial^2 f_p}{\partial x^2}(a_p, b_p) & \dfrac{\partial^2 f_p}{\partial x \partial y}(a_p, b_p) \\ \dfrac{\partial^2 f_p}{\partial x \partial y}(a_p, b_p) & \dfrac{\partial^2 f_p}{\partial y^2}(a_p, b_p) \end{pmatrix}.$$

Since (a_p, b_p) is an interior maximum and $f_p \in \mathcal{C}^2(\mathbb{R}^2)$, we have $\nabla f_p(a_p, b_p) = 0$, and the second-order Taylor expansion of f_p at (a_p, b_p) reads, for every direction $\xi \in \mathbb{R}^2$ and every t small enough so that $(a_p, b_p) + t\xi \in \mathcal{D}$:

$$\begin{aligned} f_p((a_p, b_p) + t\xi) &= f_p(a_p, b_p) + t\,\nabla f_p(a_p, b_p) \cdot \xi + \frac{t^2}{2}\,\xi^\top H_p(a_p, b_p)\,\xi + o(t^2) \\ &= f_p(a_p, b_p) + \frac{t^2}{2}\,\xi^\top H_p(a_p, b_p)\,\xi + o(t^2). \end{aligned}$$

Because $f_p((a_p,b_p)+t\xi)-f_p(a_p,b_p)\leq 0$, dividing by $t^2/2>0$ and letting $t\to 0$ yields

$$\forall \xi\in\mathbb{R}^2,\quad \xi^\top H_p(a_p,b_p)\,\xi\leq 0,$$

i.e., the Hessian matrix is always negative semi-definite at interior maximum points as (a_p,b_p). In particular, its two eigenvalues are non-positive, hence its trace is non-positive as well. But the trace of the Hessian is exactly the Laplacian:

$$\operatorname{Tr} H_p(a_p,b_p)=\Delta f_p(a_p,b_p)=\Delta f(a_p,b_p)+\frac{4}{p}=\frac{4}{p}>0,$$

which is a contradiction. (Taking $\xi=(1,0)$ and $\xi=(0,1)$ gives back the two inequalities of the first method.)

Question 3 Show that there exists $(a,b)\in\mathcal{C}$ such that

$$f(a,b)=\max_{\overline{\mathcal{D}}} f(x,y). \tag{4}$$

Solution : *The sequence $(a_p,b_p)_{p\geq 1}$ is included in the compact set $\mathcal{C}$. So there exists a subsequence $(a_{\varphi(p)},b_{\varphi(p)})_{p\geq 1}$ which converges to some $(a,b)\in\mathcal{C}$. For every $(x,y)\in\mathcal{D}$, we have $f_p(x,y)\xrightarrow{p\to+\infty} f(x,y)$. We also have the following inequality for every $p\geq 1$:*

$$f_{\varphi(p)}(x,y)\leq f(a_{\varphi(p)},b_{\varphi(p)})+\frac{r^2}{\varphi(p)}.$$

Then, by passing to the limit, we get

$$f(x,y)\leq f(a,b)$$

for every $(x,y)\in\mathcal{D}$. We have used the continuity of f.

Question 4 Show that if f and g are two harmonic functions on $\mathbb{R}^2$ equal on a circle $\mathcal{C}$, then they are equal on the entire disk inside the circle.

Solution : *We assume that the circle is centered at 0 so that we can keep the notation above. $f-g$ is harmonic, so $\max_{\mathcal{D}} f(x,y)-g(x,y)\leq 0$. Similarly, $g-f$ is harmonic, so we have the reversed inequality, which gives the result.*

Exercise 2: Resolution using Fourier's series

Let $L>0$ and $f:[0,L]\to\mathbb{R}$ be a continuous function. We consider the following problem:

$$\begin{cases} -u''=f \text{ on } (0,L),\\ u(0)=a,\\ u(L)=b, \end{cases} \tag{5}$$

where $a,b\in\mathbb{R}$.

We aim to simplify the previous problem by reducing it to a simpler one: the case with homogeneous Dirichlet boundary conditions, i.e.:

$$\begin{cases} -v''=f \text{ on } (0,L),\\ v(0)=0,\\ v(L)=0, \end{cases} \tag{6}$$

As in the lectures, we look for solutions u and v which are in $\mathcal{C}^2([0,L],\mathbb{R})$.

Question 1 Assume there exists a solution v to eq. (6). Define a function $h : [0, L] \to \mathbb{R}$ such that $u = v + h$ is a solution to eq. (5).

Solution : *If u and v are solutions to eq. (5) and eq. (6), respectively, then $h = u - v$ must satisfy the following problem:*

$$\begin{cases} h'' = 0 \text{ on } (0, L), \\ h(0) = a, \\ h(L) = b. \end{cases} \tag{7}$$

The solution is the affine function $h(x) = \dfrac{b-a}{L}x + a$.

Question 2 Show that if $(\lambda, w) \in \mathbb{R} \times \mathcal{C}^2([0, L])$ is a solution to the following problem (with w a non-zero function):

$$\begin{cases} \text{Find } (\lambda, w) \in \mathbb{R}_+^* \times \mathcal{C}^2([0, L]) \text{ such that} \\ -w'' = \lambda w \text{ on } (0, L), \\ w(0) = 0, \\ w(L) = 0, \end{cases} \tag{8}$$

then $(\lambda, \mu w)$ is also a solution for every $\mu \in \mathbb{R}$. Find all the solutions of the previous problem with w such that $\displaystyle\int_0^L w^2 = 1$.

Solution : *Since taking the second derivative is a linear operation and the boundary conditions are 0, we get that for $\mu \in \mathbb{R}$, if (λ, w) is solution, then $(\lambda, \mu w)$ is also solution.*

We first notice that necessarily, $\lambda > 0$. Indeed, by multiplying the equation $-w'' = \lambda w$ by w and integrating over $x \in (0, L)$, one gets, after an integration by parts,

$$\int_0^L (w')^2 = \lambda \int_0^L w^2,$$

which shows that $\lambda \geq 0$. Moreover, if $\lambda = 0$, then $w' = 0$ and thus, thanks to the boundary conditions, $w = 0$, which is excluded. (This can also be checked directly by writing explicitly the solutions for $\lambda \leq 0$, and by checking that one cannot obtain a non-zero solution w in this case.)

Thus, we know that, for a fixed $\lambda > 0$, the solution w to (8) is of the form $w(x) = A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}x)$. Using the boundary conditions, one gets $A = 0$ and there exists $n \in \mathbb{N}^*$ such that $\sqrt{\lambda}L = n\pi$.

The set of solutions is thus $\left(\lambda_n = \frac{n^2\pi^2}{L^2},\ w_n = \pm\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)\right)_{n\geq 1}$. The normalizing constant for w_n is determined so that $\displaystyle\int_0^L (w_n)^2 = 1$.

Question 3 Let $N \in \mathbb{N}^*$ and $f(x) = \sqrt{\frac{2}{L}}\sum_{n=1}^N f_n \sin\left(\frac{n\pi x}{L}\right)$. Find the solution v to eq. (6).

Solution : *Let us denote $e_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$ for every $n \geq 1$. From the previous question, one gets that the linear space $V_N := \operatorname{span}(e_n)_{1\leq n\leq N}$ is left invariant by the linear operator $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$. Since $f \in V_N$, it is thus natural to look for $u \in V_N$.*

Let $u(x) = \displaystyle\sum_{n=1}^N u_n e_n(x) \in V_N$. From the previous question, one has

$$-u''(x) = \sum_{n=1}^N \lambda_n u_n e_n(x).$$

One thus obtains a solution to eq. (6) by simply setting

$$u_n = \frac{f_n}{\lambda_n}.$$

This is the only solution of class $\mathcal{C}^2$, since if one considers two solutions, their difference is solution to eq. (6) with $f = 0$, and thus it is zero.
Remark 1: *One can check that $(e_n)_{1\leq n\leq N}$ is actually an orthonormal basis of V_N for the scalar product $(u, v) = \int_0^L u(x)v(x)\,\mathrm{d}x$. Indeed, $\int_0^L e_n e_m = \delta_{nm}$. One can thus recover the formula for u_n by multiplying the equation $-u'' = f$ by e_n and integrating by parts.*
Remark 2: *One can actually make sense of this procedure to build solutions even if $N = \infty$, using the notion of Fourier's series. This will be explained in forthcoming lectures on analysis and PDEs.*

Exercise 3: Green's Function for the Dirichlet Problem in Dimension 1

Question 1: A fundamental solution of the Laplace operator is defined as a function $\Phi : \mathbb{R} \to \mathbb{R}$ that is in $\mathcal{C}^2(\mathbb{R}^*, \mathbb{R})$ and satisfies, for all $x \in \mathbb{R}^*$,

$$\Phi''(x) = 0,$$

and such that for all $\varphi \in \mathcal{C}_c^2(\mathbb{R})$, the following identity holds:

$$-\int_{-\infty}^{\infty} \Phi(x)\varphi''(x)\,\mathrm{d}x = \varphi(0).$$

Find an expression for a fundamental solution of the form $\Phi(x) = v(|x|)$, where $v : \mathbb{R}_+ \to \mathbb{R}$ is to be determined.
Solution : *The fundamental solution Φ is of the form $\Phi(x) = v(|x|)$, where $v : \mathbb{R}_+ \to \mathbb{R}$ is a function that is twice differentiable over $\mathbb{R}_+^*$. For all $x \neq 0$, we have:*

$$\Phi'(x) = v'(|x|)\frac{x}{|x|}, \quad \Phi''(x) = v''(|x|).$$

Thus, for all $x \neq 0$, $v''(|x|) = 0$, which implies that there exist constants $a, b \in \mathbb{R}$ such that

$$\Phi(x) = v(|x|) = a|x| + b.$$

To determine the values of a and b, let $\varphi \in \mathcal{C}_c^2(\mathbb{R})$. Then:

$$-\int_{-\infty}^{\infty} \Phi(x)\varphi''(x)\,\mathrm{d}x = -a\int_{-\infty}^{\infty} |x|\varphi''(x)\,\mathrm{d}x - b\int_{-\infty}^{\infty} \varphi''(x)\,\mathrm{d}x.$$

We observe that:

$$\int_{-\infty}^{\infty} \varphi''(x)\,\mathrm{d}x = \lim_{R\to\infty} [\varphi'(R) - \varphi'(-R)] = 0.$$

For the first term:

$$\int_{-\infty}^{\infty} |x|\varphi''(x)\,\mathrm{d}x = \int_0^{+\infty} x\varphi''(x)\,\mathrm{d}x - \int_{-\infty}^{0} x\varphi''(x)\,\mathrm{d}x.$$

Consider the first integral:

$$\int_0^{+\infty} x\varphi''(x)\,\mathrm{d}x = [x\varphi'(x)]_0^{+\infty} - \int_0^{+\infty} \varphi'(x)\,\mathrm{d}x = \varphi(0).$$

Similarly, for the second integral:

$$\int_{-\infty}^{0} x\varphi''(x)\,\mathrm{d}x = [x\varphi'(x)]_{-\infty}^{0} - \int_{-\infty}^{0} \varphi'(x)\,\mathrm{d}x = -\varphi(0).$$

Hence:

$$\int_{-\infty}^{\infty} |x|\varphi''(x)\,\mathrm{d}x = 2\varphi(0).$$

We can therefore choose the fundamental solution as:

$$\Phi(x) = -\frac{|x|}{2},$$

but any function of the form $\Phi(x) = -\dfrac{|x|}{2} + b$ *with* $b \in \mathbb{R}$ *is also a fundamental solution.*

Question 2: Consider the Dirichlet problem on the interval $(0, L)$ with $L > 0$ and $a, b \in \mathbb{R}$:

$$\begin{cases} -u''(x) = f(x) \text{ for } x \in (0, L), \\ u(0) = a, \quad u(L) = b. \end{cases} \tag{9}$$

Derive the expression of the Green's function such that the solution of the Dirichlet problem (9) is given by:

$$u(x) = \int_0^L G(x, y) f(y)\,\mathrm{d}y - \partial_y G(x, L) b + \partial_y G(x, 0) a.$$

Recall that for the Dirichlet problem on the bounded domain $\Omega \subset \mathbb{R}^d$

$$\begin{cases} -\Delta u(x) = f(x) & \text{for } x \in \Omega, \\ u(x) = g(x) & \text{for } x \in \partial\Omega, \end{cases}$$

the Green's function $G : (x, y) \in \Omega^2 \mapsto G(x, y) \in \mathbb{R}$ satisfies:

$$u(x) = \int_\Omega G(x, y) f(y)\,\mathrm{d}y - \int_{\partial\Omega} \partial_n G(x, y) g(y)\,\sigma(\mathrm{d}y).$$

Solution : *Let* u *be a solution of the Dirichlet problem (9), which we assume to be twice differentiable. For all* $x \in (0, L)$*, we have:*

$$u'(x) - u'(0) = -\int_0^x f(t)\,\mathrm{d}t,$$

$$u(x) = a + u'(0)x - \int_0^x \int_0^t f(s)\,\mathrm{d}s\,\mathrm{d}t.$$

We simplify the double integral into a single integral:

$$\int_0^x \int_0^t f(s)\,\mathrm{d}s\,\mathrm{d}t = \int_0^x \int_s^x f(s)\,\mathrm{d}t\,\mathrm{d}s = \int_0^x (x - s) f(s)\,\mathrm{d}s.$$

Since $u(L) = b$*, we have:*

$$b = a + Lu'(0) - \int_0^L (L - s) f(s)\,\mathrm{d}s,$$

$$u'(0) = \frac{b - a}{L} + \frac{1}{L}\int_0^L (L - s) f(s)\,\mathrm{d}s.$$

Substituting this value of $u'(0)$ into the expression of u, we obtain, for all $x \in [0, L]$:

$$u(x) = \underbrace{a\left(1 - \frac{x}{L}\right) + b\frac{x}{L}}_{\text{boundary terms}} + \underbrace{\frac{x}{L}\int_0^L (L-y)f(y)\,\mathrm{d}y - \int_0^x (x-y)f(y)\,\mathrm{d}y}_{\text{terms depending on } f}.$$

The idea is now to gather all the terms depending on f *and to write them as a single integral of the form $\int_0^L G(x,y)f(y)\,\mathrm{d}y$: the kernel G obtained in this way is precisely the Green's function. To do so, we first write both integrals over the same interval $(0, L)$ using an indicator function:*

$$\int_0^x (x-y)f(y)\,\mathrm{d}y = \int_0^L \mathbb{I}\{y \le x\}(x-y)f(y)\,\mathrm{d}y,$$

so that the terms depending on f read

$$\int_0^L \left[\frac{x}{L}(L-y) - \mathbb{I}\{y \le x\}(x-y)\right] f(y)\,\mathrm{d}y =: \int_0^L G(x,y)f(y)\,\mathrm{d}y.$$

Since f is arbitrary, this identifies G: for $y > x$, the bracket equals $\frac{x}{L}(L-y)$, and for $y \le x$,

$$\frac{x}{L}(L-y) - (x-y) = \frac{xL - xy - xL + yL}{L} = y\left(1 - \frac{x}{L}\right).$$

Thus, the solution of the Dirichlet problem (9) is given by:

$$u(x) = a\left(1 - \frac{x}{L}\right) + b\frac{x}{L} + \int_0^L G(x,y)f(y)\,\mathrm{d}y,$$

where $G(x,y) = \mathbb{I}\{y \le x\}y\left(1 - \frac{x}{L}\right) + \mathbb{I}\{y > x\}\frac{x}{L}(L-y)$.
We express the Green's function using the fundamental solution of the previous question. We have for all $(x,y) \in (0,L)^2$:

$$\begin{aligned} G(x,y) &= \mathbb{I}\{y \le x\}y + \mathbb{I}\{y > x\}x - \frac{xy}{L} \\ &= \min\{x,y\} - \frac{xy}{L} \\ &= -\frac{|x-y|}{2} + \frac{x+y}{2} - \frac{xy}{L} \\ &= \Phi(x-y) - \left(\frac{xy}{L} - \frac{x+y}{2}\right). \end{aligned}$$

Furthermore, we confirm that the function $h_x(y) = \frac{xy}{L} - \frac{x+y}{2}$ is indeed a corrector function satisfying

$$\begin{cases} -h_x''(y) = 0, \\ h_x(0) = -\frac{x}{2} = \Phi(x), \\ h_x(L) = \frac{x-L}{2} = \Phi(x-L). \end{cases}$$

Exercise 4: Green's Function for the Neumann problem in dimension 1

Question 1: Consider the Neumann problem on the interval $(0, L)$ with $L > 0$ and $a', b' \in \mathbb{R}$:

$$\begin{cases} -u''(x) = f(x) \text{ for } x \in (0,L), \\ u'(0) = a', \quad u'(L) = b'. \end{cases} \tag{10}$$

What are the necessary conditions on f, a', b' for the Neumann problem (10) to admit a solution in $\mathcal{C}^2((0,L)) \cap \mathcal{C}^1([0,L])$? Show that if these conditions are satisfied, the solution is uniquely defined up to an additive constant.

Solution : *Let u be a solution of the Neumann problem (10), which we assume to be twice differentiable. We have:*

$$u'(x) = a' - \int_0^x f(t)\,\mathrm{d}t,$$
$$u'(L) = a' - \int_0^L f(t)\,\mathrm{d}t = b'.$$

Thus, if (10) admits a solution, then a', b', f must satisfy the necessary condition:

$$\int_0^L f(t)\,\mathrm{d}t = a' - b'.$$

Under this condition, we can express the solution of the Neumann problem (10) as:

$$\begin{aligned} u(x) &= u(0) + a'x - \int_0^x \int_0^t f(s)\,\mathrm{d}s\,\mathrm{d}t, \\ &= u(0) + a'x - \int_0^x (x-s) f(s)\,\mathrm{d}s, \end{aligned}$$

where $u(0)$ is an arbitrary constant. Thus, if u is a solution of (10), then for all $c \in \mathbb{R}$, $u+c$ is also a solution of (10). Therefore, there are infinitely many solutions.

Question 2: Define the function

$$N : (x,y) \in (0,L)^2 \mapsto \frac{x^2+y^2}{2L} - \max\{x,y\} + \frac{L}{3}.$$

Under the conditions established in the previous question, prove that any solution u of (10) satisfies:

$$\forall x \in [0,L], \quad u(x) = \frac{1}{L}\int_0^L u(y)\,\mathrm{d}y + \int_0^L N(x,y) f(y)\,\mathrm{d}y + b' N(x,L) - a' N(x,0). \tag{11}$$

Solution : *We first verify that the function*

$$v(x) = \int_0^L N(x,y) f(y)\,\mathrm{d}y + b' N(x,L) - a' N(x,0)$$

satisfies the boundary conditions. Notice that, for all $y \in [0,L]$ and $x \neq y$,

$$\partial_x N(x,y) = \frac{x}{L} - \mathbb{I}\{x > y\}.$$

Hence, for all $x \in [0,L]$

$$\begin{aligned} v'(x) &= \int_0^L \left(\frac{x}{L} - \mathbb{I}\{x>y\}\right) f(y)\,\mathrm{d}y + b'\frac{x}{L} - a'\left(\frac{x}{L} - 1\right) \\ &= \frac{x}{L}\left(\int_0^L f(y)\,\mathrm{d}y + b' - a'\right) - \int_0^x f(y)\,\mathrm{d}y + a' \\ &= -\int_0^x f(y)\,\mathrm{d}y + a', \end{aligned}$$

We have indeed that:

$$v'(0) = a', \quad v'(L) = b',$$
$$-v''(x) = f(x).$$

Thus, for any solution u of (10), there exists $c_u \in \mathbb{R}$ such that for all $x \in [0, L]$, $u(x) = c_u + v(x)$. To determine c_u, we integrate u over $(0, L)$:

$$\frac{1}{L}\int_0^L u(y)\,\mathrm{d}y = c_u + \frac{1}{L}\int_0^L v(y)\,\mathrm{d}y.$$

Notice that for all $y \in [0, L]$,

$$\int_0^L N(x,y)\,\mathrm{d}x = \frac{L^2}{6} + \frac{y^2}{2} - \int_0^y y\,\mathrm{d}x - \int_y^L x\,\mathrm{d}x = \frac{L^2}{2} + \frac{y^2}{2} - y^2 + \frac{y^2}{2} - \frac{L^2}{2} = 0.$$

Hence,

$$c_u = \frac{1}{L}\int_0^L u(y)\,\mathrm{d}y.$$

Remark: the Green's function of the Neumann problem can be expressed with the fundamental solution and the Neumann corrector function:

$$\forall (x,y) \in [0,L]^2, \quad N(x,y) = -\frac{|x-y|}{2} - \frac{x+y}{2} + \frac{x^2+y^2}{2L} + \frac{L}{3}.$$

Exercise 5: Numerical Resolution of the Laplace equation on a square

The goal of this exercise is to compute a numerical solution to the following problem:

$$\begin{cases} -\Delta u = f \text{ in } \Omega, \\ u = g \text{ on } \partial\Omega, \end{cases} \tag{12}$$

where $\Omega = (0,1) \times (0,1)$, $f \in \mathcal{C}^2(\Omega)$, and $g \in \mathcal{C}^0(\partial\Omega)$. To achieve this, let $1 \leq I \in \mathbb{N}$ and $h = \frac{1}{I}$. We define a Cartesian grid $(x_{ij})_{0 \leq i,j \leq I}$ by $x_{ij} = (\frac{i}{I}, \frac{j}{I})$. $(I+1)^2$ is therefore the number of points in the grid, and h is the size of each cell in the mesh.

Part 1

Question 1 Let $h > 0$ and $u \in \mathcal{C}^4([0,1])$. Show that there exists $C > 0$ such that for every $x \in [h, 1-h]$,

$$\left| \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} - u''(x) \right| \leq Ch^2. \tag{13}$$

Solution : *By the Taylor-Lagrange inequality, we have for $M = \sup_{x \in [0,1]} |u^{(4)}(x)|$ that*

$$\left| u(x+h) - u(x) - hu'(x) - \frac{h^2}{2}u''(x) - \frac{h^3}{6}u^{(3)}(x) \right| \leq \frac{M}{24}h^4, \tag{14}$$

and

$$\left| u(x-h) - u(x) + hu'(x) - \frac{h^2}{2}u''(x) + \frac{h^3}{6}u^{(3)}(x) \right| \leq \frac{M}{24}h^4. \tag{15}$$

Because $u^{(4)}$ is continuous on $[0,1]$, it is bounded by some constant M. We then obtain the result by the triangular inequality with $C = M/12$.

Question 2: Using the previous question, find a way to approximate the value of $-\Delta u$ at x_{ij} for every $1 \leq i, j \leq I-1$ using the values of u on the grid $(x_{ij})_{0 \leq i,j \leq I}$. Then, provide the full discretized problem using this method.

Solution : *We introduce the notations $u_{ij} = u(x_{ij})$, $g_{ij} = g(x_{ij})$, and $f_{ij} = f(x_{ij})$. Recall that $-\Delta u = -\partial_x^2 u - \partial_y^2 u$. Thus, we can approximate $-\Delta u$ as*

$$-\Delta u = -\frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2} + O\left(h^2\right). \tag{16}$$

The full discretized problem is then:

$$\begin{cases} -\dfrac{U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1} - 4U_{i,j}}{h^2} & = f_{ij}, \\ U_{0,j} & = g_{0j}, \\ U_{I,j} & = g_{I,j}, \\ U_{i,0} & = g_{i,0}, \\ U_{i,I} & = g_{i,I}, \end{cases} \tag{17}$$

for all $1 \leq i, j \leq I-1$, where U_{ij} aims to approximate $u(x_{ij})$ for every $0 \leq i, j \leq I$.

Part 2

In this section, we consider the case $f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$ and $g(x, y) = 0$.

Question 3 What is the true solution in this case?

Solution : *Because f can be written as the product of a function in the variable x and a function in the variable y, and because of the symmetry of x and y, the solution can be sought as $u(x, y) = g(x)g(y)$. Finally, one gets for $x = y$, $-2g(x)g''(x) = 2\pi^2 \sin(\pi x) \sin(\pi x)$. So g should be equal to $g(x) = \sin(\pi x)$. Finally, one can check that the solution is given by $u(x, y) = \sin(\pi x) \sin(\pi y)$.*

Question 4 To verify that our method is correct, we want to compare our numerical solution to the true solution. Check that the rate of convergence of the method is 2. What is the expression of the error used?

Solution : *The expression of the error is:*

$$Error(I) = \sqrt{\sum_{i,j=1}^{I-1} \frac{1}{I^2} |U_{ij} - u_{ij}|^2}. \tag{18}$$

Note that we consider only the interior points of the domain because no error occurs on the edges. This error is an approximation of the L^2-norm defined by $\left(\int_0^1 \int_0^1 |u(x, y) - U(x, y)|^2 \, \mathrm{d}x \, \mathrm{d}y \right)^{1/2}$ using the left Riemann sum (so we stop the sum at $I-1$ and not I!). The x-axis is $\log\left(\frac{1}{I}\right)$, and the y-axis is $\log(Error(I))$. Indeed, if one wants to check the rate of convergence, i.e., the number α in $Error(I) \sim \frac{1}{I^\alpha}$, then with such axes, one simply reads the slope of the curve: $\log(Error(I)) \sim -\alpha \log(I)$.

Part 3

In this section, we consider the case $f(x, y) = 0$ and $g(x, y) = xy$.

Question 5 What is the true solution in this case?

Solution : *The solution is* $u(x, y) = xy$.

Question 6 What is the rate of convergence of our method? Explain.

Solution : *There is no rate of convergence: the error is very small and remains roughly the same regardless of the number of points in the grid. All derivatives of order higher than* 2 *are* 0 *for this solution. As a result, the Taylor series used in Question 1 is exact, and so is the discretized method: the numerical solution is exact!*

Part 4

Question 7 What happens if, instead of $-\Delta u = f$, one considers $-\Delta u + c(x, y)u = f$ in the equation, where c is continuous and non-negative? Code and plot the rate of convergence of the method for this case. If there is no analytical solution, the errors can be computed using a reference solution, i.e., a solution computed with a large number of points.

Solution : *Compared with Question 2, only one term changes. At each interior point* x_{ij}, *the term* $c(x_{ij})\, u(x_{ij})$ *does not need to be approximated: we simply multiply the unknown* U_{ij} *by the value* $c_{ij} = c(x_{ij})$. *The discretized problem becomes*

$$\begin{cases} -\dfrac{U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1} - 4U_{ij}}{h^2} + c_{ij}\, U_{ij} = f_{ij}, & 1 \le i, j \le I - 1, \\ U_{ij} = g_{ij}, & \text{on the boundary of the grid.} \end{cases} \tag{19}$$

In the code, this amounts to adding c_{ij} *to the diagonal of the matrix of the linear system.*

What rate of convergence should we expect? *The only approximation in the scheme is still the one of Question 1, which makes an error of order* h^2. *The new term* $c\,u$ *adds no approximation error, since it is computed exactly at the grid points. We therefore expect the method to remain of order* 2.

Why do we need $c \ge 0$***?*** *Adding a non-negative number to the diagonal makes the system "more stable": it can always be solved, and small errors are not amplified. If* c *were negative, things could go wrong. For instance, with* $c \equiv -2\pi^2$, $f = 0$ *and* $g = 0$, *both* $u = 0$ *and* $u(x, y) = \sin(\pi x)\sin(\pi y)$ *are solutions: the problem no longer has a unique solution.*

Numerical experiments. *We take* $c(x, y) = 50(x^2 + y^2)$, $g = 0$, *and two test cases.*

- Case (a): known solution. *We choose the solution first,* $u(x, y) = \sin(\pi x)\sin(\pi y)$, *and deduce the right-hand side:* $f = -\Delta u + c\,u = (2\pi^2 + c)\,u$. *The error is computed as in Question 5.*
- Case (b): unknown solution. *We take* $f(x, y) = 2\pi^2 \sin(\pi x)\sin(\pi y)$. *There is no explicit formula for* u, *so we replace it by a reference solution computed on a very fine grid* ($I_{ref} = 1024$), *and compare at the points shared by both grids.*

Figure 1 shows the results for $I = 8, 16, 32, 64, 128$. *In both cases, the points lie on a straight line of slope* 2: *each time* h *is divided by* 2, *the error is divided by* 4. *The method is still of order* 2.

Remark. *The reference solution is itself only an approximation. It is reliable only if its grid is much finer than those we test, here* $I_{ref} = 1024$ *against* $I \le 128$.

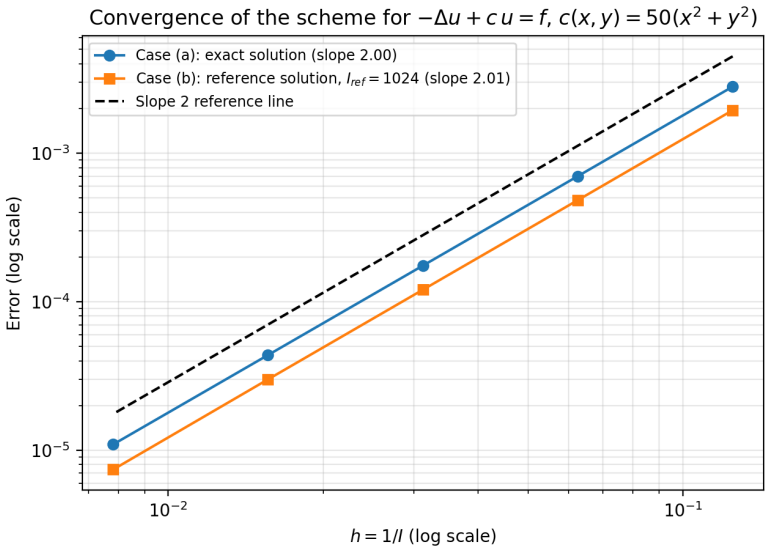


Figure 1: Error as a function of h in log-log scale for $-\Delta u + c\,u = f$.